

## K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

## End Semester Exam

Oct – Nov 2015

Max. Marks:100

Class: SE

Name of the Course: Applied Mathematics – III

Course Code : UCEC301

Duration: 3hr.

Semester: III

Branch: Computer Engineering

## Instructions:

- (1) All Questions are Compulsory  
 (2) Figures to right indicate full marks. Each sub-question has equal marks.

| Question NO. | Questions                                                                                                                                                                                                                      | Marks |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Q.1          | <b>A Attempt any THREE of the following.</b>                                                                                                                                                                                   | 18    |
|              | (i) Find Laplace Transform of (i) $e^t \sin^3 t$ (ii) $\int_0^t \frac{\sin u}{u} du$                                                                                                                                           |       |
|              | (ii) Find Inverse Laplace Transform of (i) $\frac{s^2}{(s+1)^3}$ (ii) $\log \left( \sqrt{\frac{s^2+1}{s^2}} \right)$                                                                                                           |       |
|              | (iii) Find Laplace Transform of $f(t) = \begin{cases} 1, & 0 \leq t < a \\ -1, & a \leq t < 2a \end{cases}$<br>Where $f(t)$ is a periodic function with period $2a$ .                                                          |       |
|              | (iv) Evaluate $\int_0^\infty e^{-t} \sin\left(\frac{t}{2}\right) \sinh\left(\frac{\sqrt{3}}{2}t\right) dt$                                                                                                                     |       |
|              | <b>B Attempt any ONE of the following.</b>                                                                                                                                                                                     | 7     |
|              | (i) State Convolution Theorem and using Convolution theorem find $L^{-1} \left[ \frac{1}{(s+3)(s^2+2s+2)} \right]$                                                                                                             |       |
|              | (ii) Solve the following Differential Equation using Laplace Transform $(D^2 + D)y = t^2 + 2t$ , $y(0) = 4$ , $y'(0) = -2$                                                                                                     |       |
| Q.2          | <b>A Attempt anyTHREE of the following.</b>                                                                                                                                                                                    | 18    |
|              | (i) Find the Fourier series of $f(x) = \begin{cases} 1, & 0 \leq x \leq \pi \\ 2 - \frac{x}{\pi}, & \pi \leq x \leq 2\pi \end{cases}$                                                                                          |       |
|              | (ii) Find the Fourier series of $f(x) = x - x^2$ in $(-1,1)$ .                                                                                                                                                                 |       |
|              | (iii) Find Half Range Cosine series of $f(x) = \sin x$ in $(0,\pi)$                                                                                                                                                            |       |
|              | (iv) Find the Fourier Sine Transform of $f(x) = e^{-x}$ , $x \geq 0$ and hence deduce that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi}{2} e^{-m}$ , $m > 0$                                                         |       |
|              | <b>B Attempt anyONE of the following.</b>                                                                                                                                                                                      | 7     |
|              | (i) Find the Fourier series of $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$ State the value of $f(x)$ at $x = 0$ and hence deduce that $\sum_{n=1}^\infty \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$ |       |
|              | (ii) Find the Complex form of Fourier Series of $f(x) = 2x$ in $(0,2\pi)$                                                                                                                                                      |       |

|     |          |                                                                                                                                                                                                                                                                          |           |
|-----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Q.3 | <b>A</b> | <b>Attempt any THREE of the following.</b>                                                                                                                                                                                                                               | <b>18</b> |
|     | (i)      | If $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ , find the Eigen values of<br>(i) $A^3 - 3A^2 + A$ (ii) $Adj A$ (iii) $A^{-1}$                                                                                                             |           |
|     | (ii)     | If $A = \begin{bmatrix} \frac{\pi}{2} & \pi \\ 0 & \frac{3\pi}{2} \end{bmatrix}$ , find $\sin A$ and $\cos A$                                                                                                                                                            |           |
|     | (iii)    | Show that $A = \begin{bmatrix} 7 & 4 & -1 \\ 4 & 7 & -1 \\ -4 & -4 & 4 \end{bmatrix}$ is derogatory and write its minimal Polynomial.                                                                                                                                    |           |
|     | (iv)     | Find the Eigen values and the Eigen Vectors of $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$                                                                                                                                                   |           |
|     | <b>B</b> | <b>Attempt any ONE of the following.</b>                                                                                                                                                                                                                                 | <b>7</b>  |
|     | (i)      | Show that $A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$ is Diagonalisable. Write the Transforming matrix M and the Diagonal form D.                                                                                                        |           |
|     | (ii)     | Verify Cayley Hamilton Theorem for $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence evaluate $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$                                                                                    |           |
| Q.4 | <b>A</b> | <b>Attempt any THREE of the following.</b>                                                                                                                                                                                                                               | <b>18</b> |
|     | (i)      | Prove that $\nabla \cdot \left\{ \frac{f(r)}{r} \vec{r} \right\} = \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)]$ .<br>Hence show that $\nabla \cdot \left\{ \frac{\log r}{r} \vec{r} \right\} = \frac{1}{r} [1 + 2 \log r]$ .                                                   |           |
|     | (ii)     | Find the directional derivative of $\phi = e^{2x} \cos(yz)$ at $(0,0,0)$ in the direction of the tangent to the curve $x = a \sin t, y = a \cos t, z = at$ at $t = \frac{\pi}{4}$ .                                                                                      |           |
|     | (iii)    | Find the angle between the surfaces $ax^2 + y^2 + z^2 - xy = 1$ and $bx^2y + y^2z + z = 1$ at $(1,1,0)$ .                                                                                                                                                                |           |
|     | (iv)     | Evaluate by Green's Theorem $\int_C [(3x^2 - 8y^2)dx + (4y - 6xy)dy]$ where C is the boundary of the region bounded by $y = \sqrt{x}$ and $y = x$ .                                                                                                                      |           |
|     | <b>B</b> | <b>Attempt any ONE of the following.</b>                                                                                                                                                                                                                                 | <b>7</b>  |
|     | (i)      | Find a,b,c if $\vec{F} = (x + 2y + az)\vec{i} + (bx - 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is irrotational. Find its Scalar Potential and the work done in moving a particle in this field from $(1,2,-4)$ to $(3,3,2)$ along the straight line joining these points. |           |
|     | (ii)     | Use Gauss's Divergence Theorem to evaluate $\iint_S \vec{N} \cdot \vec{F} ds$ where $\vec{F} = 4xi - 2y^2j + z^2k$ and S is the region bounded by $x^2 + y^2 = 4, z = 0, z = 3$                                                                                          |           |